MIL-L-49148A(ER)
1 October 1980
SUPERSEDING
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1 February 1979

MILITARY SPECIFICATION

LENS ASSEMBLY, OBJECTIVE VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

This specification is approved for use by USAERADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Objective Lens Assembly, herein known as objective lens, which is part of Viewer, Driver's Night Vision, AN/VVS-2(V).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-Pallet, Materials Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	-Strapping, Steel, and Seals
NNN-P-40	-Paper, Lens
PPP-B-566	-Boxes, Folding, Paperboard
PPP-B-585	-Boxes, Wood, Wirebound
PPP-B-601	-Boxes, Wood, Cleated-Plywood
PPP-B-621	-Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-Boxes, Shipping, Fiberboard
PPP-B-640	-Boxes, Fiberboard, Corrugated, Triple Wall
PPP-C-843	-Cushioning Material, Cellulosic
PPP-S-760	-Strapping, Nonmetallic (and Connectors)
PPP-T-97	-Tape, Packaging/Industrial, Filament Reinforced

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783 by using the self addressed Standarization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	-Preservation-Packaging, Methods of
MIL-B-117	-Bags, Sleeves and Tubing - Interior Packaging
MIL-P-11268	-Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	-Marking of Electronic Items
MIL-O-13830	-Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-Marking for Shipment and Storage
MIL-STD-150	-Photographic Lenses
MIL-STD-171	-Finishing of Metal and Wood Surfaces
MIL-STD-252	-Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and other Devices.
MIL-STD-810	-Environmental Test Methods
MIL-STD-1188	-Commercial Packaging of Supplies and Equipment

DRAWINGS

ELECTRONICS COMMAND

DL SM-B-771334	-Barrel Assembly, Lens, Objective

(Copies of Specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The objective lens assembly is mounted to and is part of the Viewer, Driver's Night Vision AN/VVS-2(V).

3.2 Construction. The objective lens assembly shall be constructed in accordance with DL SM-B-771334 and as specified herein.

3.3 First article. The supplier shall furnish and test the objective lens as specified in the contract or purchase order (see 6.2).

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3.4 Parts and materials. Parts and materials shall be as specified herein and shown on applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.5 Performance characteristics.

3.5.1 Effective focal length. The objective lens shall have an Effective Focal Length (EFL) of 33.6, ±0.5 millimeters (mm).

3.5.2 Field of view. The field of view shall be not less than 44 degrees for a 25 mm diameter format.

3.5.3 Linear distortion. The objective lens shall have between 9 percent and 12 percent barrel distortion.

3.5.4 Modulation Transfer Function (MTF) on axis. The MTF shall be not less than the values specified in Table I.

TABLE I. Objective lens modulation transfer function.

Test Frequency (cycles/millimeter) on the Image Intensifier	Modulation Transfer (Percent)
10	84
20	70
30	55
40	40

3.5.5 T-number. The T-number shall be no greater than 1.25.

3.5.6 Veiling glare. The veiling glare shall be no greater than 2.5 percent.

3.5.7 Sealing. The objective lens shall be assembled and sealed in an atmosphere of dry nitrogen containing 1 mole-percent helium. The helium leak rate of the sealed objective lens shall be not greater than 9.9×10^{-4} cm^3/sec.

3.5.8 Flange focal length. The flange focal length shall be not less than 0.039 inches (0.991 mm) when measured from the 1.5-inch diameter central area of retainer (SM-C-771354).

3.6 Operation and environment.

3.6.1 High temperature. The objective lens shall operate at a temperature up to +52°C. The objective lens shall withstand storage temperature of +68°C without damage or degradation of performance (see 6.3).

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3.6.2 Low temperature. The objective lens shall operate at a temperature down to -54°C. The objective lens shall withstand storage temperature of -57°C without damage or degradation (see 6.3).

3.6.3 Temperature shock. The objective lens shall withstand cycles of sudden changes in temperature between -54°C and +68°C without damage or degradation of performance (see 6.3).

3.6.4 Vibration. The objective lens shall withstand vibration equivalent to that encountered in tracked vehicles during 5000 miles of operation without damage or degradation of performance (see 6.3).

3.6.5 Shock. The objective lens shall withstand basic shock of 40g's for 18 milliseconds (ms). The objective lens shall withstand simulated weapon firing shock of 100g's for 1.5 ms on half sine pulse without degradation of performance (see 6.3).

3.7 Treatment and painting. Unless otherwise specified, treatment and painting shall conform to MIL-STD-171.

3.8 Marking. Unless otherwise specified, the objective lens shall be marked and identified per MIL-M-13231.

3.9 Workmanship. Standards of workmanship shall be such that components will meet all requirements of this specification and any referenced specifications or drawings where inspected in accordance with section 4. Parts and assemblies shall exhibit none of the defects of type 9 or type 11 as categorized in MIL-STD-252.

3.9.1 Cleanliness. The objective lens shall meet the cleanliness requirement for optical systems as specified in MIL-O-13830.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract the contractor may use his own or any or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (4.3)

b. Quality conformance inspection (4.4)

c. Inspection of packaging (4.7)

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4.3 First article inspection.

4.3.1 Inspection. Each first article objective lens shall be inspected in accordance with Table II, either in process or where assembled, as required. Presence of one or more defects shall be cause for rejection of that objective lens and may be cause for rejection of first article.

TABLE II. Inspection.

Defects	Requirement paragraph
Parts and materials not as specified	3.4
Treatment and painting	3.7
Identification or special marking missing or illegible	3.8
Workmanship	3.9
Cleanliness	3.9.1

4.3.2 Tests. Following successful completion of the inspections specified in 4.3.1, four first article objective lenses shall be subjected to all tests listed in Table III. Tests may be performed in any order. Failure of any inspection shall be cause for rejection of that assembly and may be cause for rejection of first article.

TABLE III. First article inspection.

Inspection	Requirement Paragraph	Test Paragraph
Effective focal length	3.5.1	4.5.1
Field of view	3.5.2	4.5.2
Linear distortion	3.5.3	4.5.3
Modulation transfer function	3.5.4	4.5.4
T-number	3.5.5	4.5.5
Veiling glare	3.5.6	4.5.6
Sealing	3.5.7	4.5.7
Flange focal length	3.5.8	4.5.8
High temperature	3.6.1	4.6.1
Low temperature	3.6.2	4.6.2
Temperature shock	3.6.3	4.6.3
Vibration	3.6.4	4.6.4
Shock	3.6.5	4.6.5

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4.4 Quality conformance inspection.

4.4.1 Examination. Each objective lens shall be examined as specified in Table II. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests.

4.4.2.1 Group A inspection. Objective lenses that have passed the examination specified in 4.4.1 shall be tested in accordance with Table IV in the order listed. Failure of any test shall be cause for rejection of that unit. Assemblies which pass Group A inspection shall be formed into lots (see 6.3).

TABLE IV. Group A inspection.

Test	Requirement Paragraph	Test Paragraph
Effective focal length	3.5.1	4.5.1
Field of view	3.5.2	4.5.2
Modulation transfer function	3.5.4	4.5.4
Sealing	3.5.7	4.5.7
Flange focal length	3.5.8	4.5.8

4.4.2.2 Group B. Group B tests shall be conducted on objective lenses selected from units which have passed the examination and the Group A test. Sampling shall be in accordance with MIL-STD-105, AQL 2.5 percent defective, inspection level S-3. Tests shall be conducted in accordance with Table V.

TABLE V. Group B Inspection.

Test	Requirement Paragraph	Test Paragraph
Linear distortion	3.5.3	4.5.3
T-number	3.5.5	4.5.5
Veiling glare	3.5.6	4.5.6

4.4.2.3 Group C. This inspection shall consist of the tests specified in Table VI and shall be performed on units that have been subjected to and have met Group A inspection. Units shall be selected in accordance with 4.4.2.3.1.

4.4.2.3.1 Sampling for Group C inspection. Three objective lenses shall be selected at random by the Government from each quality conformance lot or from 30 days production, whichever occurs first. The units shall be subjected to the inspections specified in Table VI.

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TABLE VI. Group C inspection.

Test	Requirement Paragraph	Test Paragraph
High temperature	3.6.1	4.6.1
Low temperature	3.6.2	4.6.2
Temperature shock	3.6.3	4.6.3
Vibration	3.6.4	4.6.4
Shock	3.6.5	4.6.5

4.4.2.3.2 Group C failures. Actions required in the event of a Group C failure shall be as specified in the contract or purchase order (see 6.2).

4.5 Test procedures. Unless otherwise specified all tests shall be performed at ambient temperature. Temperature chambers shall maintain specified temperatures within $\pm 1.4°C$ (see 6.3).

4.5.1 Effective focal length. The effective focal length of the objective lens shall be measured in accordance with the procedures of 5.1.2.2 of MIL-STD-150. Failure to meet the requirements of 3.5.1 shall constitute failure of this test.

4.5.2 Field of view. The field of view of the objective lens shall be measured in accordance with MIL-STD-150. Failure to meet the requirements of 3.5.2 shall constitute failure of this test.

4.5.3 Linear distortion. The linear distortion of the objective lens shall be measured in accordance with MIL-STD-150, and the percent distortion shall be calculated with the following formula:

$$\text{Percent distortion} = \frac{F_A - F_I}{F_I} \times 100$$

Where:

F_A = Actual image height measured (mm)

F_I = Ideal image height (mm)

$= EFL \times \tan \theta$

and:

EFL = Effective focal length (see 4.5.1)

$$= \frac{F_I(\theta_1)}{\tan \theta_1}$$

where $\theta_1 = 1°$
and $F_I(\theta_1)$ is image height measured at $\theta_1 = 1°$

Failure to meet the requirements of 3.5.3 shall constitute failure of this test.

4.5.4 Modulation transfer function. The objective lens shall be tested on axis with a sine wave pattern setup that utilizes an S-20ER response

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photomultiplier conforming to Figure 1 and a tungsten illumination source operated at 2854 K, ±200 K color temperature. (If a microscope objective is used to relay the image of the objective lens, the numerical aperture (NA) of the microscope objective must be equal to or greater than 0.65). The pattern-collimator combination must allow the objective lens to be tested either continously or at 8 distinct evenly distributed points from 0 to 40 line pairs per millimeter (lp/mm) including 0 lp/mm. The test shall be repeated with the objective rotated 90 degrees about the optical axis. Failure to meet the requirements of 3.5.4. shall constitute failure of this test.

4.5.5 T-number. The T-number of the objective lens shall be measured in accordance with Method 5 or 6 of MIL-STD-150. The light source used shall be operated at 2854 K, ±200 K color temperature. The detector used shall have an S-20ER response (see figure 1). Failure to meet the requirments of 3.5.5 shall constitute failure of this test.

4.5.6 Veiling glare. The stray light or veiling glare of the objective lens shall be measured in accordance with Method 35 of MIL-STD-150. The detector used in this procedure shall have an S-20ER response (see Figure 1). Failure to meet the requirements of 3.5.6 shall constitute failure of this test.

4.5.7 Sealing. The objective lens shall be tested for helium leakage with a Varian Model 925-40 "Porta-Test" or equivalent. A leak rate greater than that specified in 3.5.7 shall constitute failure of this test.

4.5.8 Flange focal length. The flange focal length shall be measured as the distance from the flange as described in 3.5.8 to the aerial image through the best definition. This measurement shall be performed utilizing a microscope which is aligned to the reference flange. Failure to meet the requirements of 3.5.8 shall constitute failure of this test.

4.6 Operation and environment. The objective lens shall be retested to meet the requirements of 3.5.4, 3.5.6, and 3.5.7 after all operation and environmental testing has been completed. Failure to meet these requirements shall constitute failure as defined in degradation of performance (see 6.3.6).

4.6.1 High temperature. The objective lens shall be subjected to the test of Method 501.1, Procedure 1, of MIL-STD-810, except that temperature of Steps 4 and 5 (highest operating temperature) shall be + 52°C. Failure to meet the requirements of 3.6.1 shall constitute failure of this test.

4.6.2 Low temperature. The objective lens shall be subjected to the test of Method 502.1, Procedure 1, of MIL-STD-810. The storage temperature (Step 2) shall be -57°C and shall be maintained until temperature stabilization of the test item is reached. Low operating temperature (Step 4) shall be -54°C. Failure to meet the requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Temperature shock. The objective lens shall be subjected to the test of Method 503.1, Procedure I, of MIL-STD-810. Immediately after each cold cycle, the assembly shall be examined for internal condensation. Evidence of moisture or condensation on any internal surface or failure to meet the requirements of 3.6.3 shall constitute failure of this test.

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4.6.4 Vibration. The objective lens shall be subjected to MIL-STD-810, Method 514.2, Procedure VIII, Table 514.2-6 tracked vehicles, limited to one sweep in the vertical axis only. Failure to meet requirements of 3.6.4 shall constitute failure of this test.

4.6.5 Shock. The objective lens shall be subjected to the test of MIL-STD-810, Method 516.2. Failure to meet requirements of 3.6.5 shall constitute failure of this test.

a. 40g's at 18 milliseconds, Figure 516.2-1, shall be the basic design test, Shock Procedure I.

b. Weapon firings, 100g's using procedure IV except at 1.5 ms, Figure 516.2-2.

4.7 Inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance with the requirements of Section 5 (see 6.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Objective lenses shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Objective lenses shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4. Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1. Technical literature. Each technical literature shall be unit packed method IC-1.

5.1.1.4.2 Lens assembly, objective. Each objective lens shall be individually unit packed method IA-8 as follows: wrap each unit with lens tissue conforming to NNN-P-40. Cushion each assembly by wrapping with cushioning material conforming to PPP-C-843, type II. Place the cushioned item within a barrier bag fabricated of material conforming to MIL-B-117, type II, class E, style 1. Heat seal the closure, leaving a small opening for exhausting air. Exhaust the entrapped air and complete the seal. Place the bagged item within a close-fitting box conforming to PPP-B-566, variety 2. Place the technical literature unit packed as specified in 5.1.1.4.1 on top of the contents directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

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5.1.1.4.3 Intermediate packing. Multiples of 5 not to exceed 20 objective lenses, unit packed as specified in 5.1, shall be placed within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application, unit packing and intermediate packing shall be as specified in 5.1, except that method III shall be used in lieu of method IA-8.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of objective lenses, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.2 Level B. A quantity of objective lenses, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be pressure sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial Packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

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6. NOTES

6.1 Intended use. The objective lens is used with the AN/VVS-2(V) for driver viewing during periods of darkness.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.

b. Necessary actions by contractor in the event of a Group C failure (see 4.4.2.3.2).

c. Level A, Level B, or commercial preservation and packing (see section 5).

d. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

e. If rough handling and functional tests are required.

f. Quantity and schedule for first article testing for all line items.

6.3 Definitions.

6.3.1 Damage. Damage is defined as any breakage, loosening, shift, dents, failure of any finish, hardware, connection, or component, or condensation, fogging or moisture on any internal surface.

6.3.2 Ambient temperature. Ambient temperature is defined as +23°C, ±10°C.

6.3.3 Inspection lot size. An inspection lot is defined as not more than those assemblies successfully completing Group A tests each month.

6.3.4 Environmental tests. For the purpose of this specification, environmental tests are those tests discussed in 4.6.

6.3.5 Degradation of performance. Any deterioration in quality, or failure to operate in accordance with the requirements of this specification.

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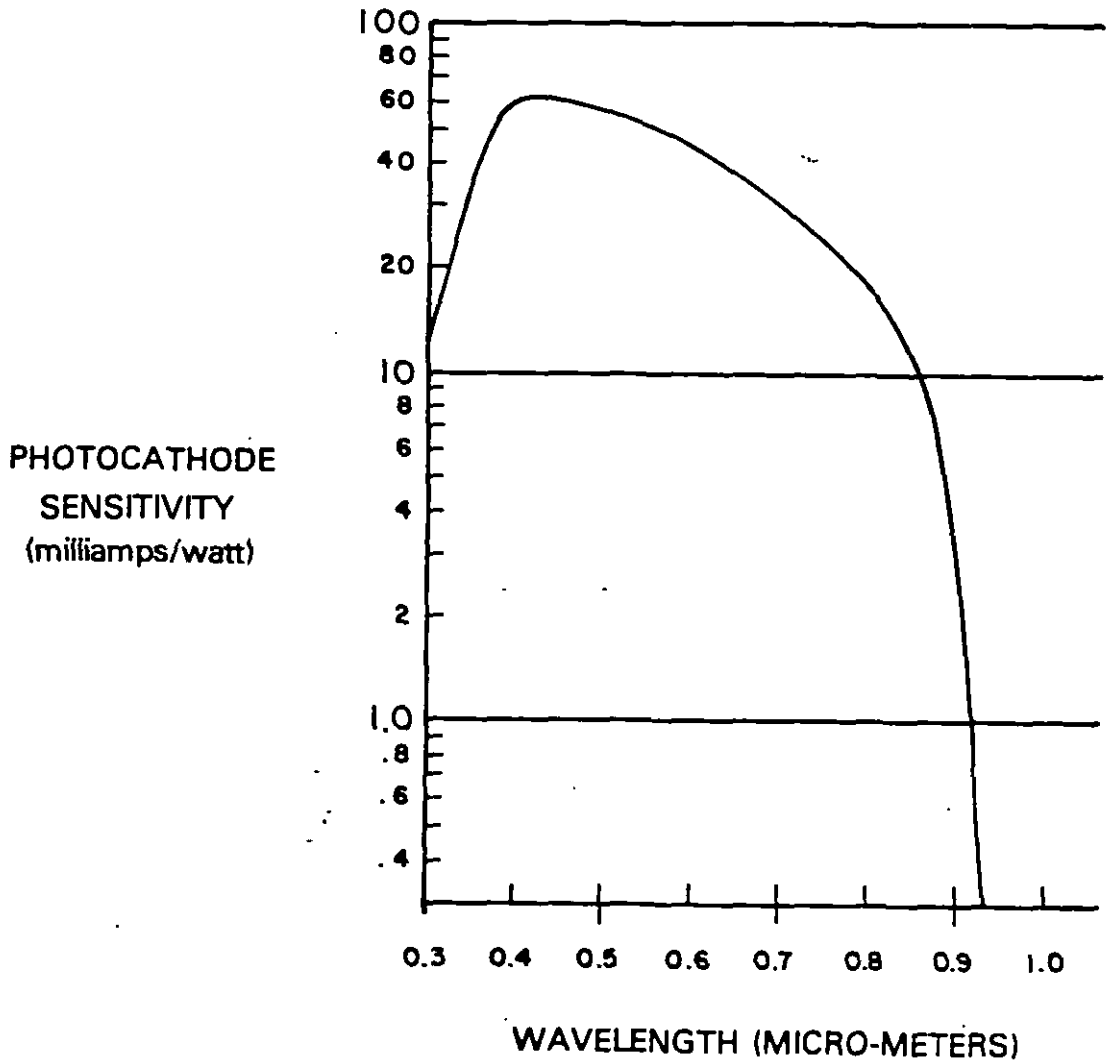


FIGURE 1. RESPONSE CURVE, S-20 EXTENDED RED PHOTOCATHODE

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